THESIS — The Fractal EBM Harness as a Path to Collective, Non-Parametric, Realtime, Private Intelligence

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The bet, in one sentence

Intelligence is **retrieval + routing + verification over a complete, live index of the world** — not weights. Scale it with *coverage* and *collective experience*, not parameters; and the marginal cost of a thought falls toward the cost of a lookup and a fold.

This is the opposite scaling law from frozen parametric LLMs, and it's what makes the four properties the user named — **lean, private, realtime, collective** — fall out of the architecture rather than being bolted on.

Why each property follows from the design

Lean / low-compute — because knowledge lives in the index, not the weights

The substrate's EBM is **non-parametric**: per-key centroids (`ebm.zig`, `E = −cosine(x, mean)`) and thin linear energy heads (`jepa.zig`, $||Wx−y||^2$), each ~1 KB, **reconstructible from the ledger** (`replayFromLedger`). The "knowledge" is not baked into billions of parameters — it's the *retrievable index* (the world, via unbrowse) plus the *collective firing history* (the ledger). A thought costs a retrieval + a centroid fold, not a forward pass through a giant net. The model doesn't *store* the world; it *addresses* it.

Always realtime — because the index updates now and the model folds on-fire

A parametric model is stale the instant its training cutoff passes. This system has no cutoff: **unbrowse indexes the live internet**, and the EBM **folds on every declare/satisfy/die** (train-on-fire). The collective's beliefs reflect the world *and its own experience* as of this moment. New fact indexed -> available immediately. New verdict fired -> the prior sharpens immediately. No retrain lag.

Private — because inference is local and you contribute verdicts, not data

Intelligence runs **locally** over your own data plus a shared index; your data stays sealed under your key (envelope encryption + the receptor grant, from the storage plan). You don't ship your data to train someone else's model — you fire locally and contribute **signed, capability-gated verdicts** to the commons. The raw stays yours; only *what worked* (attested, not the content) joins the shared prior.

Collective — because the ledger is shared memory and the graph is the mind

The ledger is a shared, append-only memory the whole swarm reads and reconstructs the EBM from. Every agent's terminal verdicts fold into the same reconstructible prior; the **independent multi-witness corroboration** requirement keeps that prior honest. No single node is "the model" — the **graph of contracts + the shared ledger-EBM + the indexed world** is the mind. Intelligence is a property of the collective, the way a market or a science is.

How the fractal EBM harness specifically helps

- **Fractal keys -> multi-scale priors for free.** The *same* atom folds across `invoke:{event}`, `:L{layer}`, `:dim:{w6h}`, `:verb:`, `:ptype:`, `:root`. One cheap operation yields a coarse-to-fine router — the collective can prioritize at every granularity without a separate model per scale.
- **Train-on-fire -> the system's own history is the training signal.** Satisfied = good, died = retry. No labels, no pipeline, no training run — the substrate learns from doing.
- **Ledger-reconstructible weights -> the model is the commons, portable and auditable.** Anyone can rebuild the prior from the shared ledger; it can be forked, inspected, and re-derived. The "weights" *are* the collective experience, in the open.

- **Energy-ranked resolution -> a thinking-budget allocator.** It routes scarce compute / spend to where history says it pays (the plan's Phase 1). This is the mechanism by which a low-compute system stays low-compute *as it scales*: it spends attention where the energy gradient is steepest.

The compounding loop (why it could grow toward "smarter than its parts")

- **Parametric AI:** intelligence ∝ params × data × compute — static, centralized, and your data trains their model.
- **This:** intelligence ∝ *index coverage* × *collective firing history* × *routing/verification quality* — live, distributed, and your data stays yours.
- As unbrowse indexes more of the world and more agents fire and corroborate, two things widen together: the *retrievable context* (what can be brought to bear) and the *sharpness of the prior* (what history says works). Each is cheap per step and never goes stale. A system whose knowledge and whose judgment both compound with use, at near-zero marginal cost, is a genuine candidate for exceeding any individual part.

THE HONEST WALL — what must be true, and where this breaks

(In the substrate's own tradition: state the limits as loudly as the vision, or it's marketing.)

- **The current energy model is shallow — a router prior, not a mind.** A cosine-to-one-centroid and a linear `w` are soft priors over "what resolved before." Real reasoning is *not* in this model today. The leap from "thin router" to "intelligence" is unbuilt, and it comes from one of two places, both unproven at this scale: richer *composed* energy models (compute cost returns), or *emergent* collective composition (that many thin routers + verification compose into reasoning). Do not mistake the scaffold for the building.
- **Retrieval ≠ reasoning.** Indexing the whole internet gives *knowledge access*, not thought. Superintelligence needs composition, planning, and verification *over* retrieved knowledge. The energy-descent-as-generation primitive (`jepa.plan`) is the seed of this, but it's linear and shallow. The reasoning layer is the hard, unsolved core.
- **"Low compute" is low *marginal*, not low *total*.** Embedding and serving retrieval over the live web at scale is real, heavy infrastructure. The compute *moves* from training to indexing/serving — it does not vanish. The honest claim is cheap-per-thought, not cheap-overall.
- **Verification is the crux, and the single biggest risk.** A collective that learns on-fire from verdicts is only as good as its verdicts. Bad or gamed verdicts don't just add noise — they *train the shared prior toward delusion*. A canonical reference anchor + independent multi-party corroboration is the designed guard, and it is load-bearing: if it fails, the collective gets confidently, collectively wrong. This is the difference between a wisdom-of-crowds and a madness-of-crowds, and it is not automatic.
- **Query privacy over a shared index is unsolved by encryption-at-rest.** Your data-at-rest is private (sealed). But *querying* a shared index can leak what you're asking. True query privacy needs PIR / local replicas / ZK — a separate, hard problem. "Private" holds for your data; not yet for your attention.
- **"Superintelligence" is a bet, not a result.** Collective systems — markets, evolution, science — *do* exceed any participant, so it isn't crazy. But whether *this* reaches "super," and not just "useful," is untested and hinges entirely on #1, #2, and #4 holding. Claim it as a *falsifiable direction*, not a destination.

The falsifiable first step (this is the discipline)

The entire thesis has a smallest testable version, and it's already the plan's **Phase 1**: does energy-ranked resolution *measurably* beat scan-order at choosing what fires next? That single question is the atom of "the collective routes its own thinking." **If the thin version doesn't help, the grand version won't** — and we'll have learned it for the cost of a benchmark instead of a cathedral. Prove the atom before believing the collective. Everything above earns its right to be built one measured gate at a time.